LAB 01: Network Modes Comparison Report

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1. Objective

This lab investigates how VirtualBox's five network adapter modes NAT, NAT Network, Bridged, Host-Only, and Internal Network affect connectivity between two virtual machines (Kali Linux and Metasploitable2), along with their reachability from the host machine and access to the internet. Each mode was configured and tested individually using live connectivity checks (ip a, ping), with results captured as screenshots rather than assumed documentation. The objective is to establish a correctly understood, deliberately isolated lab network as the foundation for all future offensive and defensive security labs, and to build a practical understanding of network isolation because misunderstanding network boundaries is one of the most common sources of both misconfigured lab environments and real-world detection blind spots.

2. Lab Environment

2.1. Host Machine:

- OS: Windows 11
- Hypervisor: Oracle VirtualBox 7.2.12

2.2.Kali Linux (Attacker/Analysis Machine):

- Version: kali-linux-2026.2-virtualbox-amd64
- RAM Allocated: 2048 MB
- CPUs: 2
- Role: Primary attacker and packet analysis machine in all future labs

2.3.Metasploitable2 (Target Machine):

- Version: Metasploitable2 (Ubuntu 8.04-based, intentionally vulnerable)
- RAM Allocated: 512 MB
- CPUs: 1
- Role: Intentionally vulnerable targets never expose this machine to a real network

3. Networking Concepts Primer

Before reading the results, these concepts must be understood clearly they are referenced throughout every mode's analysis.

3.1.What ip a show and how to read it:

ip a (short for ip address) lists every network interface on the machine along with its current IP configuration. The key lines to read are:

- eth0: the name of the first Ethernet interface. This is the "door" all network traffic passes through. A second adapter would be eth1, and so on.
- inet 10.0.2.15/24: the IPv4 address assigned to that interface. The /24 is CIDR notation meaning the first 24 bits identify the network, and the remaining 8 bits identify individual hosts. This gives 256 possible addresses (.0 to .255) on that subnet, with .0 reserved as the network address and .255 as the broadcast address.
- link/ether 08:00:27:xx:xx:xx : the MAC address (hardware-level address) of that interface. This is what ARP resolves IP addresses to. Every frame on a local network is addressed to a MAC, not an IP.
- brd ff:ff:ff:ff:ff:ff : the broadcast MAC address. Any frame sent here reaches every device on the local segment simultaneously.

3.2.Subnet mask / CIDR:

Two machines must share the same subnet to communicate directly without a router. 192.168.56.101 and 192.168.56.102 share the same /24 network (192.168.56.0) so they can communicate directly. A machine at 10.0.2.15 is on an entirely different network and would require routing to reach.

3.3.What ping does:

ping sends ICMP Echo Request packets to a target and listens for ICMP Echo Reply packets back. It tells you three things: whether a path exists to the target, how long the round trip takes (latency), and whether any packets are lost. A sub-millisecond ping to yourself is normal (loopback). A 1–10ms ping to another VM on the same host is normal. A 40–100ms ping to an internet address is normal.

3.4.What "Network is unreachable" means vs a timeout:

These are two completely different failures and it is critical to distinguish them:

- "Network is unreachable” : the sending machine's own kernel looked up its routing table and found no route to that destination at all. The packet was never sent onto the wire. This happens before any network traffic occurs.
- A timeout (no reply, but ping keeps sending) : a route exists and packets are genuinely sent, but nothing answers. The destination may not exist, may be filtering ICMP, or may be down.

4. Methodology

The following procedure was applied identically to each of the five modes:

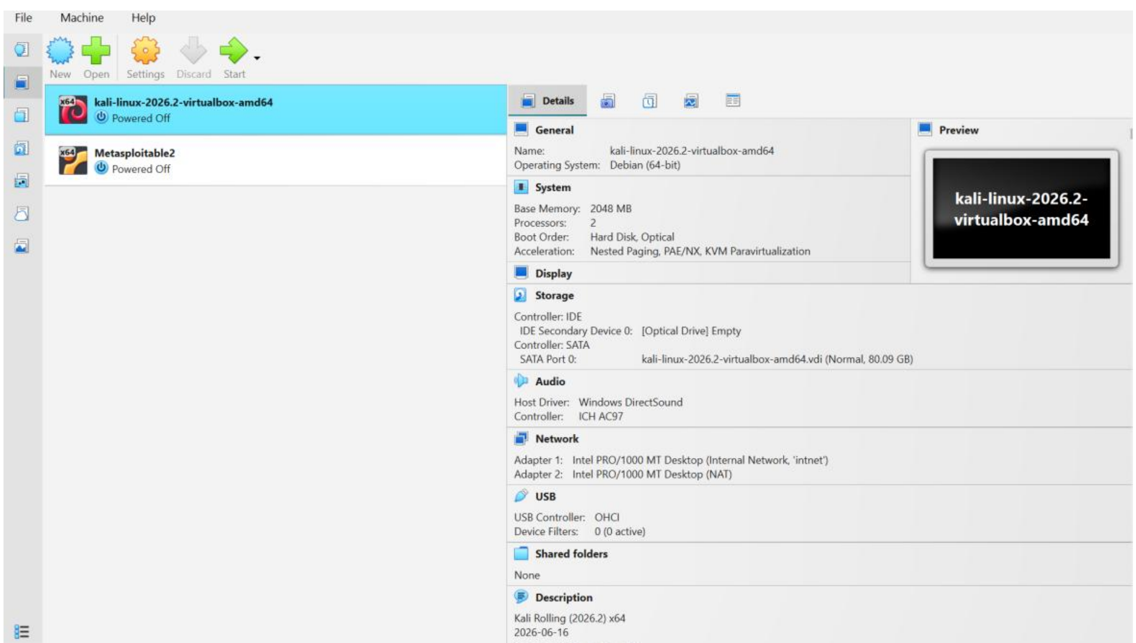
1. Power off both VMs completely (not saved state VirtualBox requires full power-off to change adapter settings).
2. Configure Adapter 1 on both Kali and Metasploitable2 to the mode under test.
3. Boot both VMs and run ip a on each to record the assigned IP address, subnet, and whether addressing was automatic or manual.
4. Test VM-to-VM connectivity in both directions using ping -c 4 <target-ip>.
5. Test internet access from each VM using ping -c 4 8.8.8.8.
6. Test host-to-VM reachability by pinging both VM IPs from Windows PowerShell.
7. Screenshot every step as evidence.

Special handling per mode is noted in that mode's section (e.g. NAT Network requires creating the network first; Internal Network requires manual static IP assignment since no DHCP server exists).

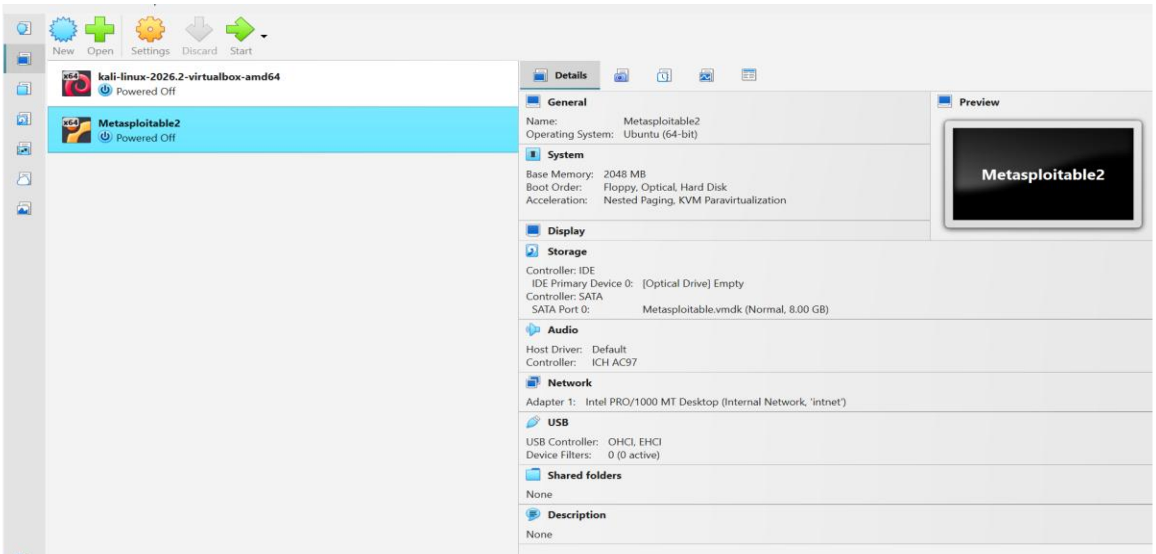
5. Setup Tasks

1. Confirm current specifications of both VMs (RAM, CPU, VirtualBox version) record these, they go in your Lab Environment section.

5.1.For kali Linux:

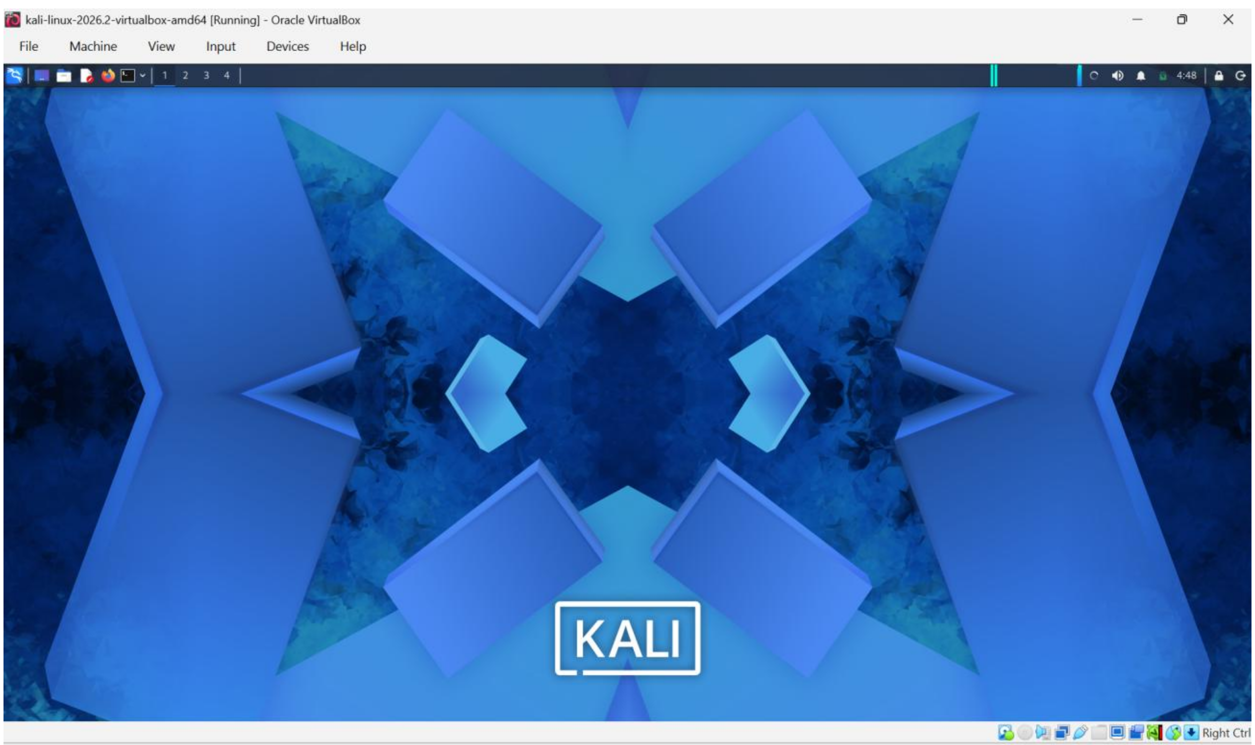


5.2.For metasploitable2:

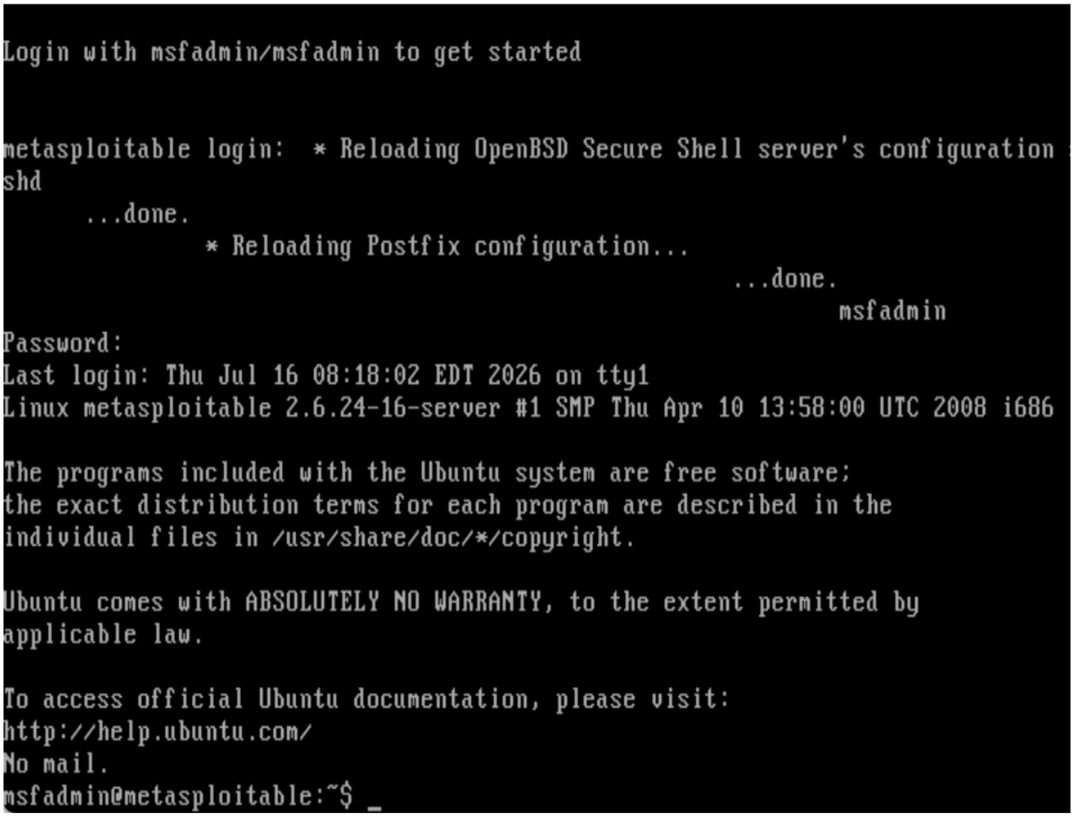


2. Make sure both VMs boot cleanly and you can log in (Kali: your user; Metasploitable2: msfadmin/msfadmin).

For kali Linux:



For metasploitable2:



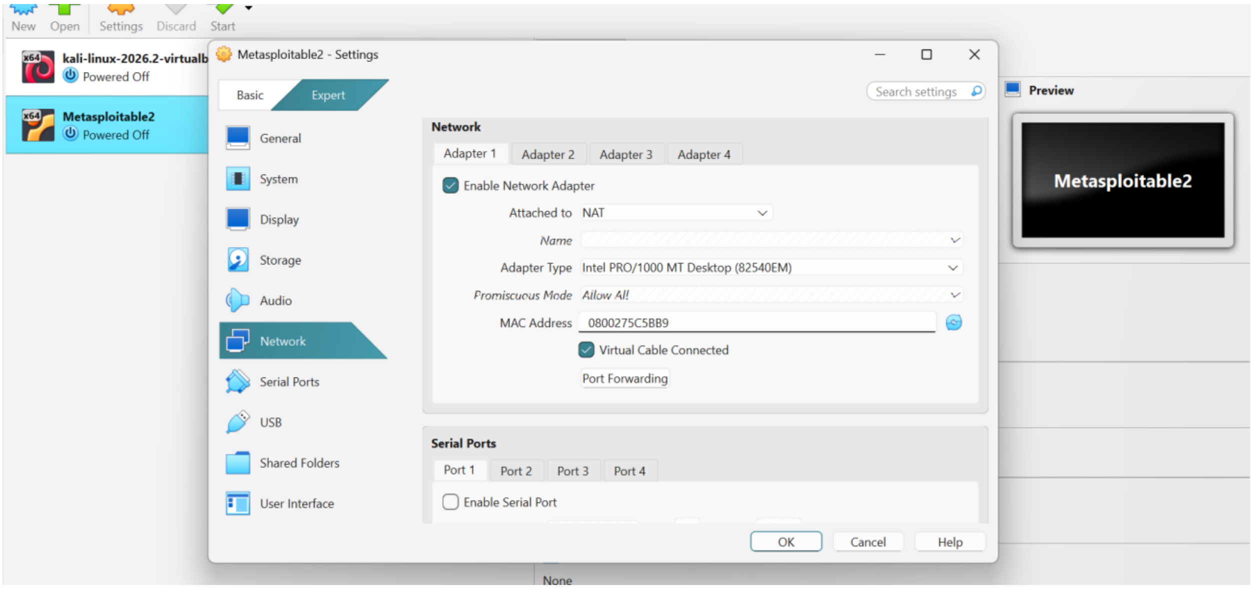
Mode 1: NAT (Network Address Translation)

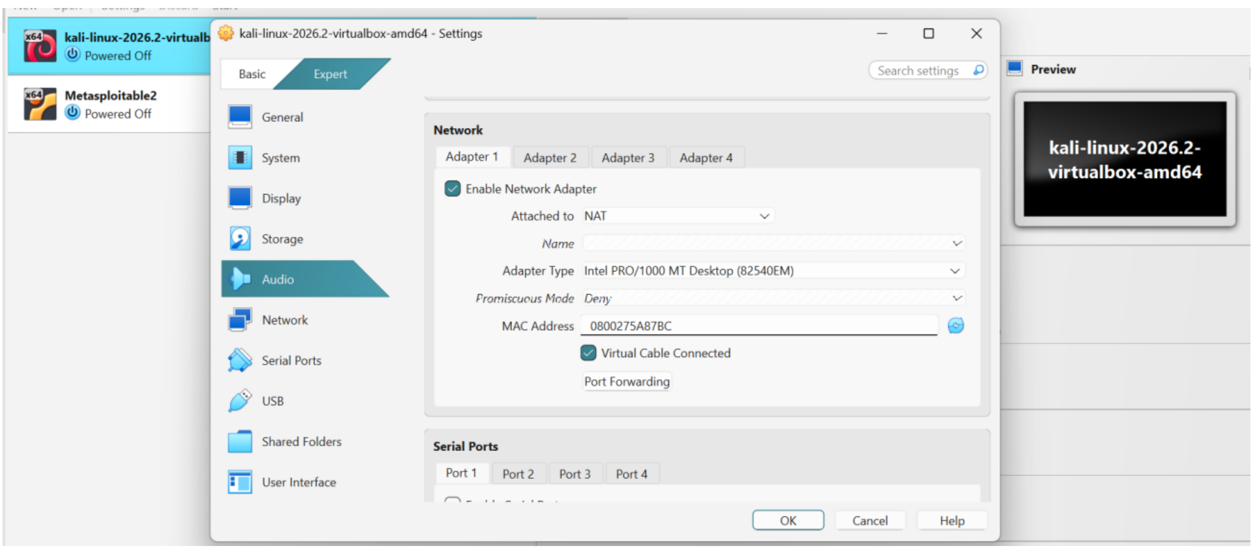
Configuration: Both Kali and Metasploitable2 set to NAT — the VirtualBox default.

Architecture:

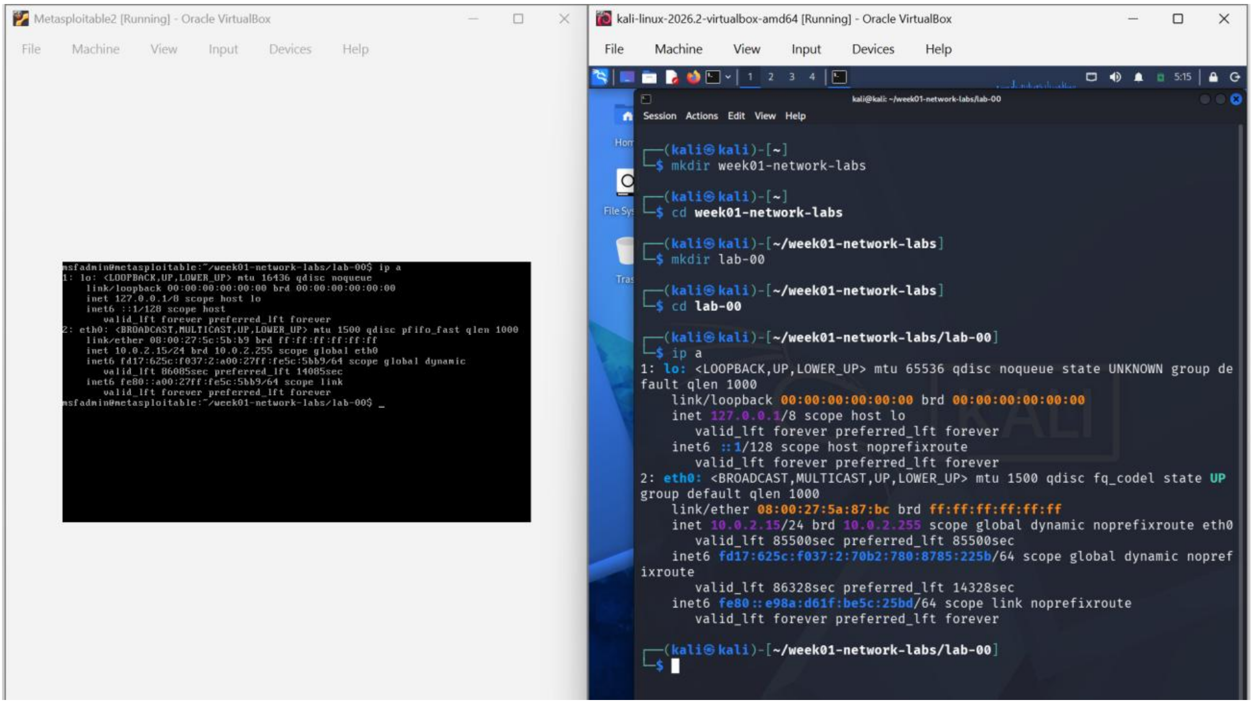
```
Internet
   │
Host (VirtualBox NAT Engine)
   │
   ├── Kali (10.0.2.15) → [isolated private bubble]
   │
   └── Metasploitable2 (10.0.2.15) → [separate isolated bubble]
```

No path between Kali and Metasploitable2


New
Open
Settings
Discard
Start
kali-linux-2026.2-virtualb
Powered Off
Metasploitable2
Powered Off
Metasploitable2 - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
NAT
Name
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Allow All
MAC Address
0800275C5BB9
Virtual Cable Connected
Port Forwarding
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
OK
Cancel
Help
Preview
Metasploitable2
None


kali-linux-2026.2-virtualb
Powered Off
Metasploitable2
Powered Off
kali-linux-2026.2-virtualbox-amd64 - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
NAT
Name
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Deny
MAC Address
0800275A87BC
Virtual Cable Connected
Port Forwarding
Serial Ports
Port 1
Port 2
Port 3
Port 4
OK
Cancel
Help
Preview
kali-linux-2026.2-
virtualbox-amd64

IP addressing check on both machines:

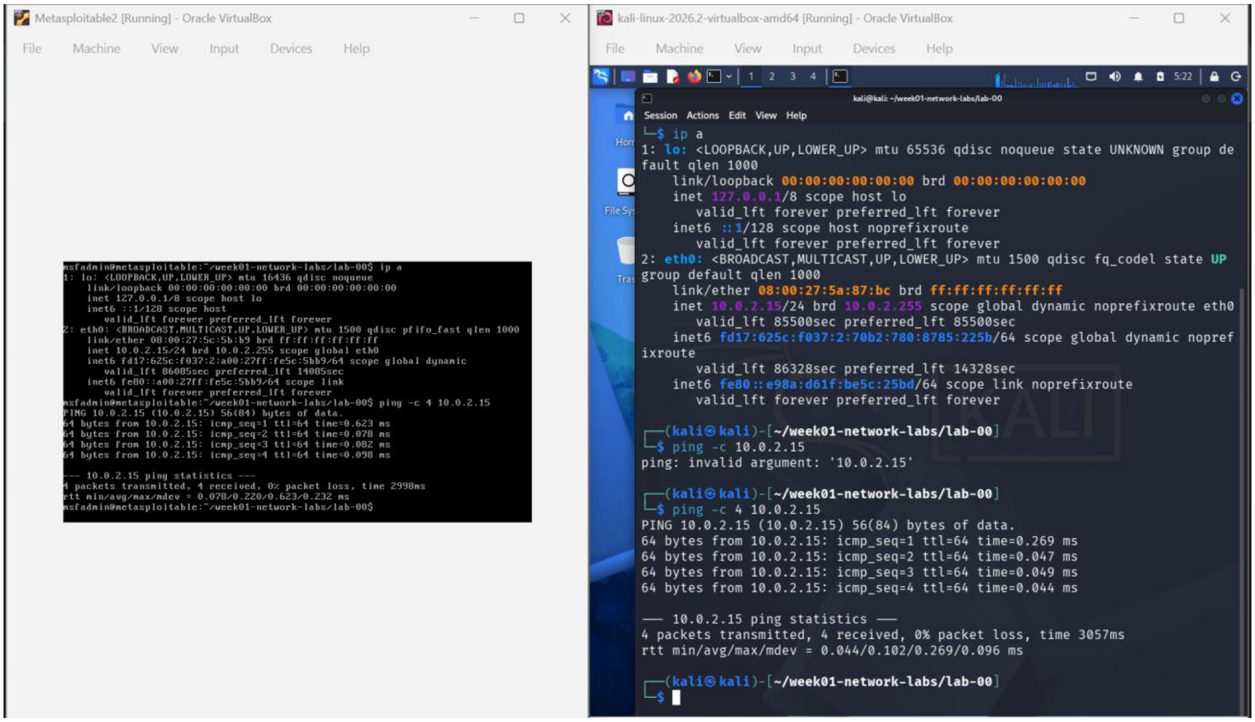


For kali linux : 10.0.2.15/24 and broadcast address : 10.0.2.255

For metasploitablw2: 10.0.2.15/24 and broadcast address : 10.0.2.255

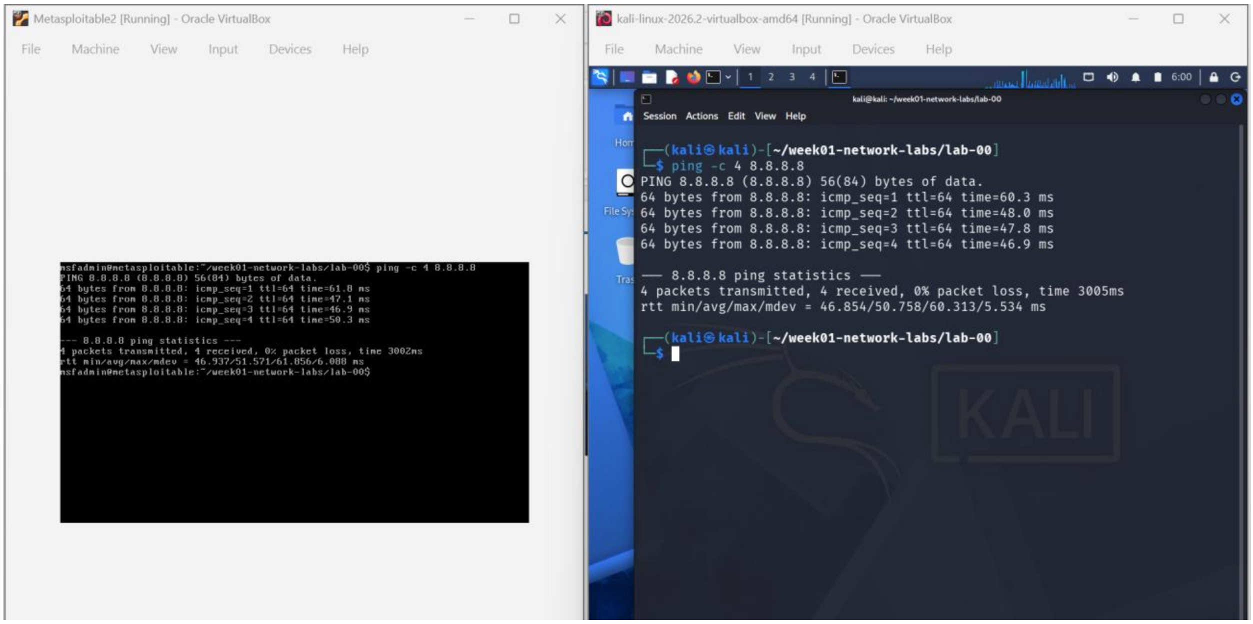
Both machines received the exact same IP address. This is not a configuration error — it is the defining behavior of NAT mode. VirtualBox creates a completely separate private virtual network for each VM individually. Each VM exists in its own isolated bubble, so two VMs can both be assigned 10.0.2.15 simultaneously without any conflict, because those addresses exist on different, unconnected network instances.

VM- TO -VM test



When Kali pings 10.0.2.15, it is pinging itself its own IP, on its own isolated network. There is no other device on Kali's NAT network. The ping "succeeds" but the sub-millisecond response time is the giveaway: the packet never left Kali's own networking stack. This is a self-loop, not genuine VM-to-VM communication.

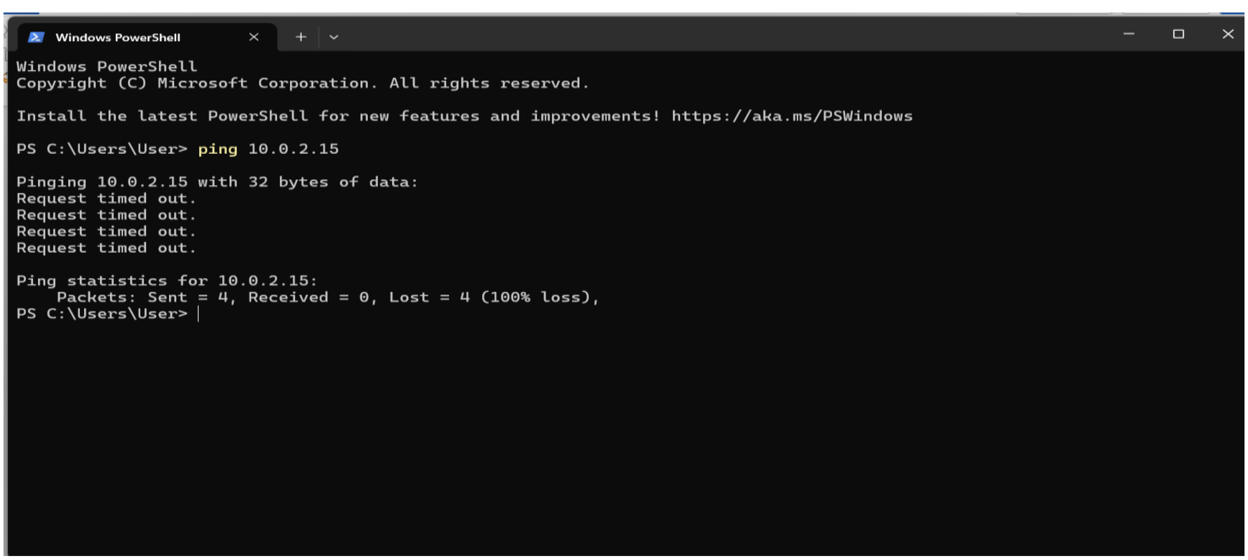
Internet test:



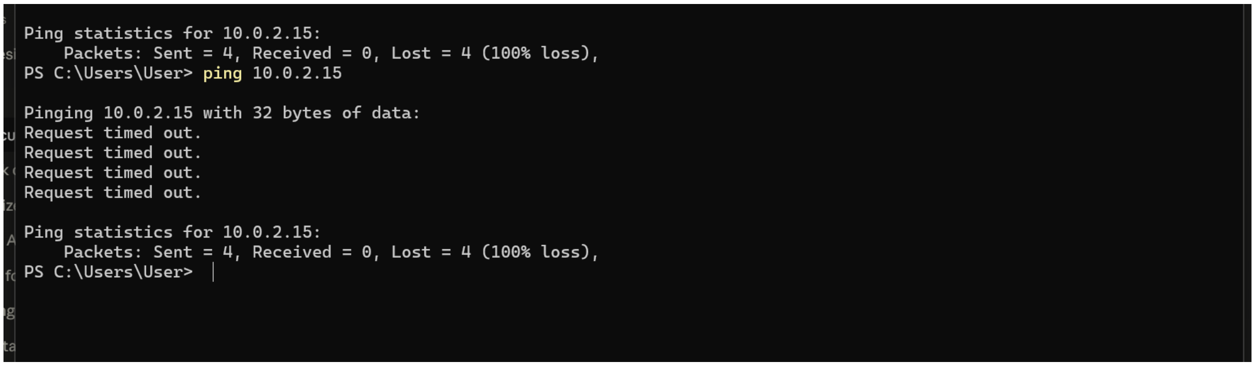
Both VMs successfully reached 8.8.8.8. VirtualBox's NAT engine intercepts all outbound traffic from each VM, rewrites the source address to the host's real IP, and forwards it through the host's actual network connection. Replies come back to VirtualBox, which forwards them to the correct VM. The VM never knows its traffic is being translated — this is how NAT works in real routers as well (your home router does the same thing for all devices on your WiFi).

Host-to-VM test:

Kali ping ip address on poweshell:



Metasploitable2 ping ip address on powershell:



```
Ping statistics for 10.0.2.15:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PS C:\Users\User> ping 10.0.2.15

Pinging 10.0.2.15 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.2.15:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PS C:\Users\User>
```

Not applicable for NAT mode. The host has no route into the VM's isolated private network. VirtualBox's NAT engine only translates outbound traffic from VMs — it does not provide a path inward from the host.

Why NAT mode exists: It is the safest and simplest default. A VM gets internet access immediately without any configuration, and it is completely unreachable from the outside. Suitable for a single VM that needs internet access and does not need to communicate with other VMs.

Why it fails for security labs: Any lab requiring two machines to interact (attacker → target) is impossible in NAT mode. The machines are on separate networks and cannot discover or reach each other.

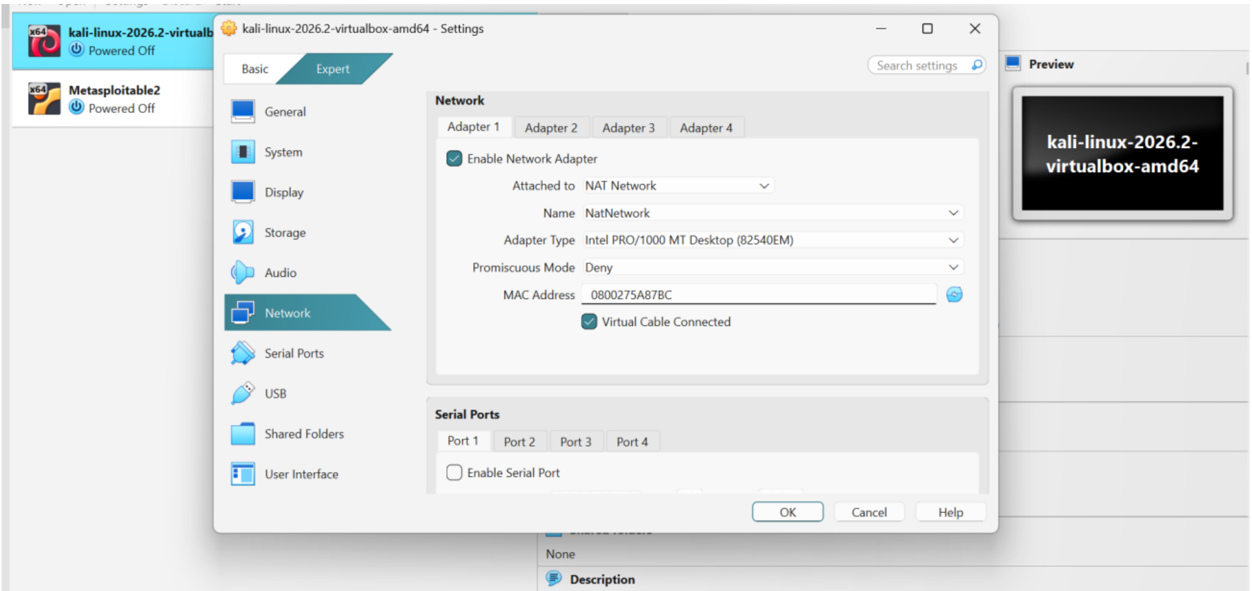
Mode 2: NAT Network

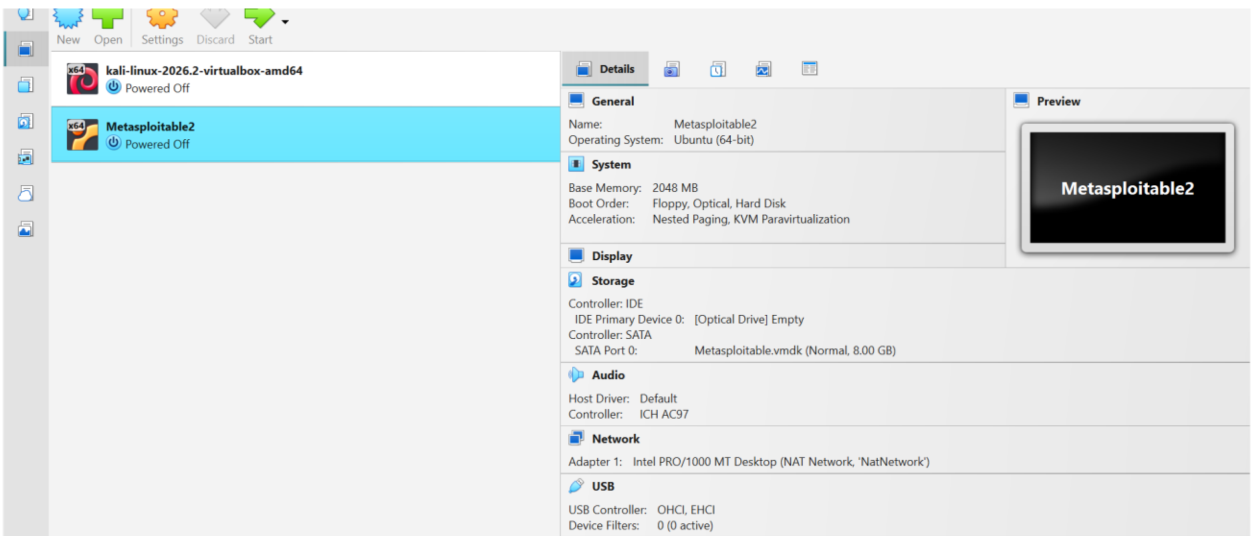
Configuration: Both VMs set to NAT Network. A NAT Network named "NatNetwork" was created first via VirtualBox Network Manager (File → Tools → Network Manager → NAT Networks → Create).

Architecture:

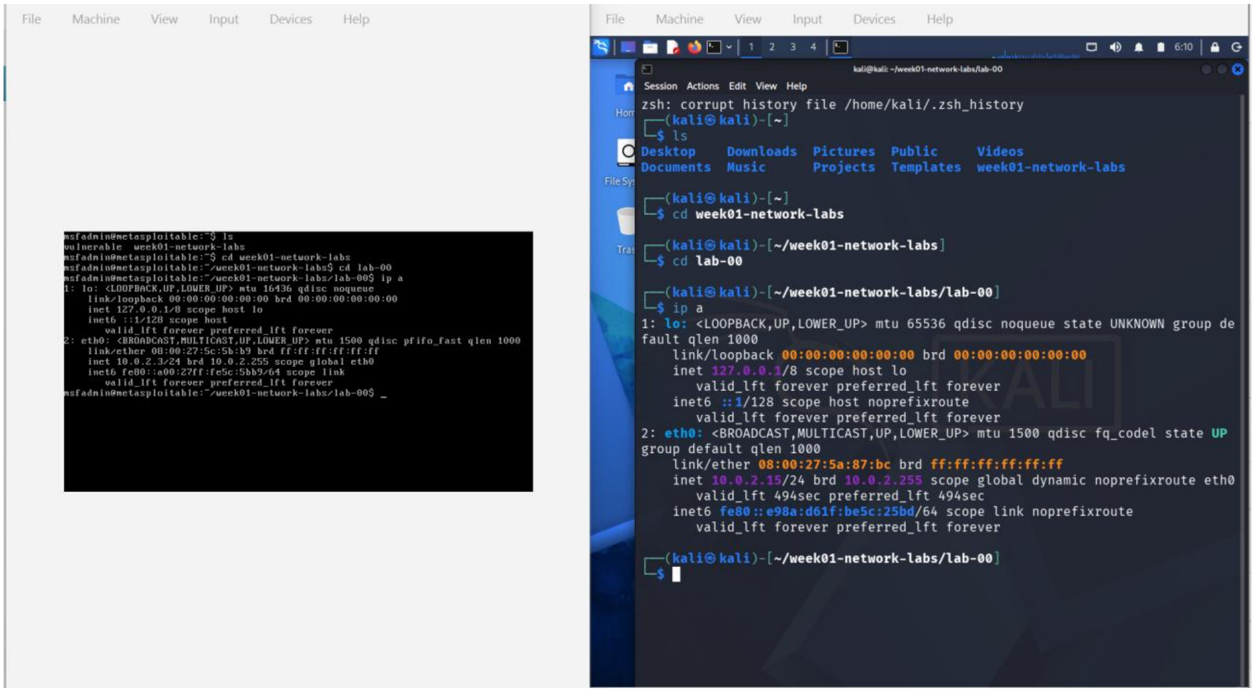
```
Internet
   |
Host (VirtualBox NAT Engine)
   |
NAT Network (10.0.2.0/24) — one shared network
   |
   ├── Kali (10.0.2.15)
   |
   └── Metasploitable2 (10.0.2.3)
```

Both VMs share one network: VM-to-VM communication + internet





IP addresses:

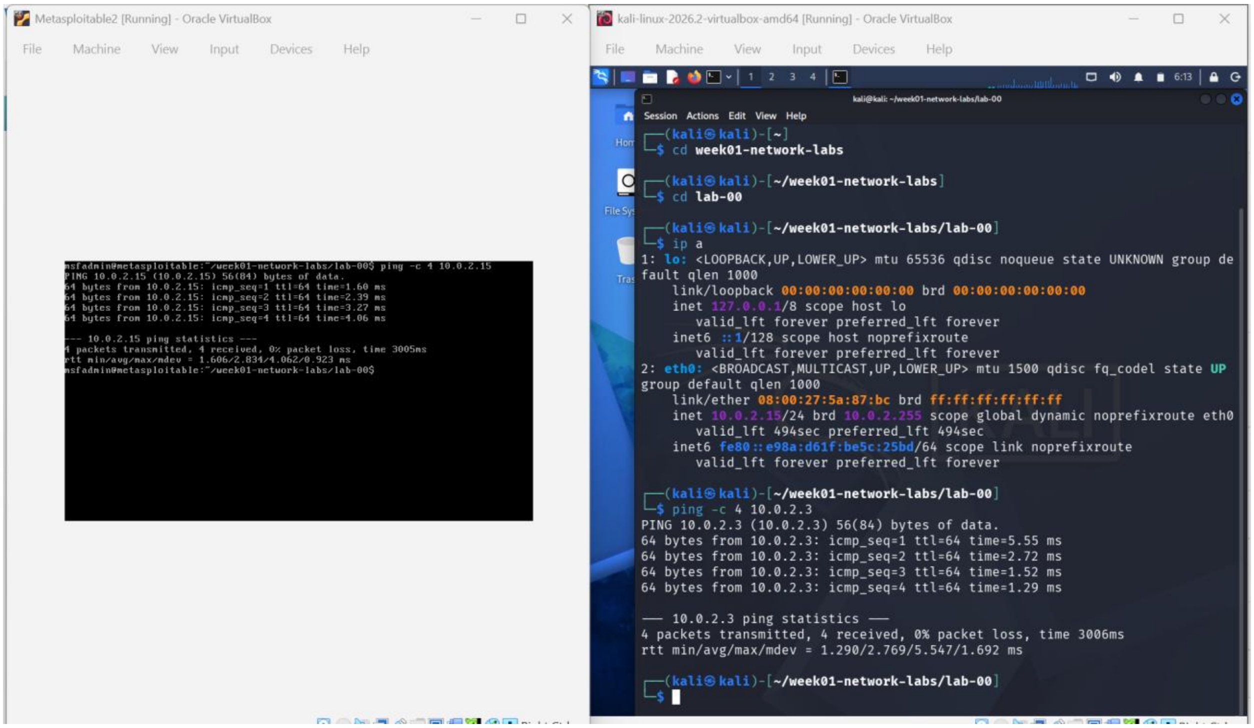


For kali linux: inet :10.0.2.15/24 broadcast address: 10.0.2.255

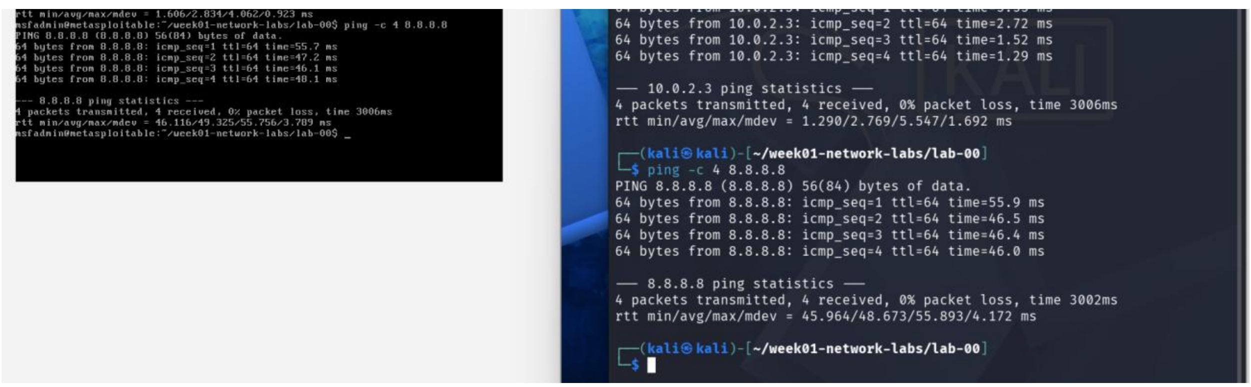
For metasploitable2: inet: 10.0.2.3/24 broadcast address: 10.0.2.255

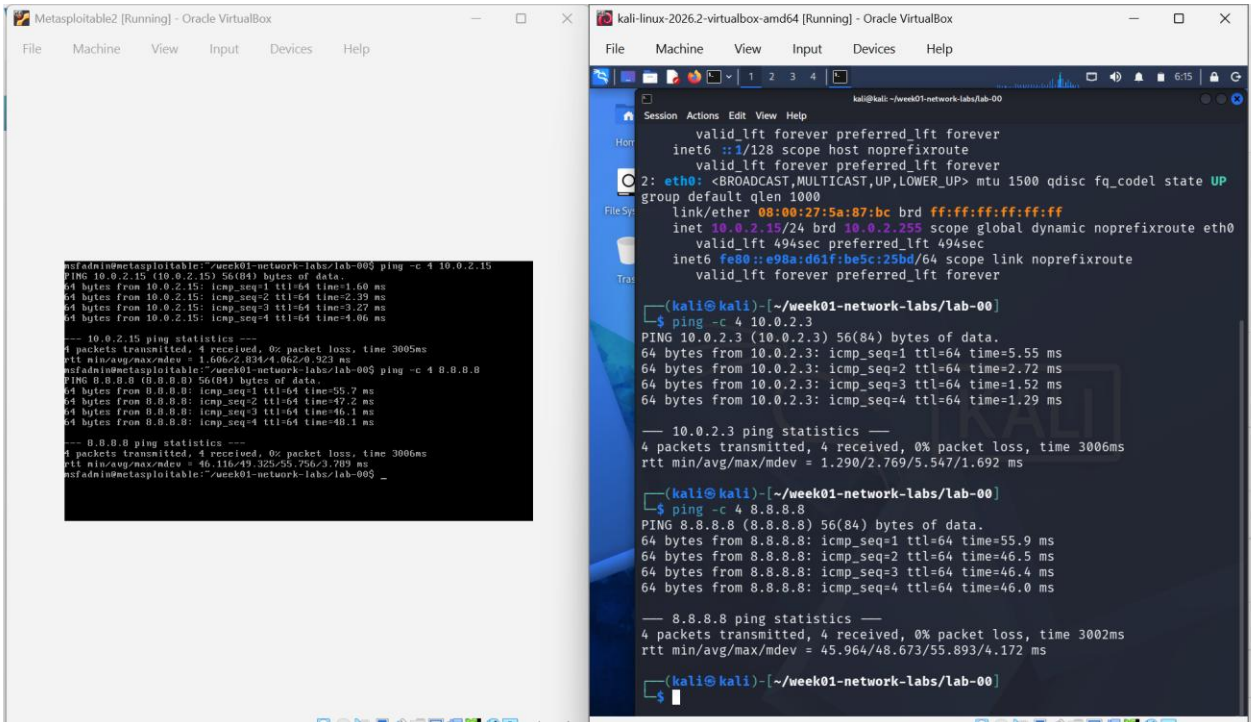
This time the addresses are different, both within the same subnet. This is because NAT Network places all VMs on a single shared virtual network with one common DHCP server. Both machines drew from the same address pool, so they received different addresses — unlike NAT mode where each had its own isolated pool and both ended up at .15.

<u>**VM-to-VM test:**</u>



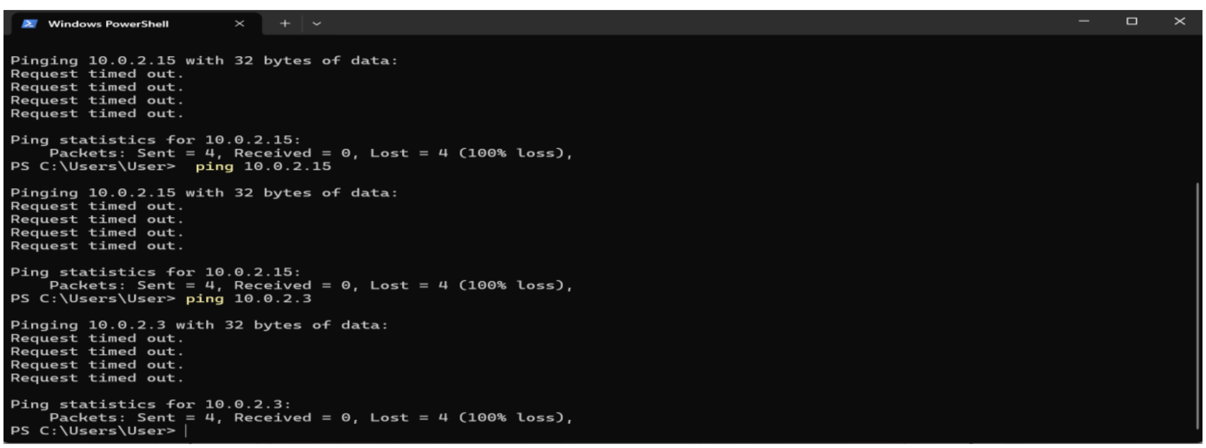
<u>**Test internet:**</u>





Communication succeeded in both directions. The latency (2–9ms) is measurably higher than the sub-millisecond self-loop seen in NAT mode this confirms genuine packets were traversing a network path between two separate machines, not looping back

Test host to VM:



Not directly applicable. The host is not a member of the NAT Network's shared subnet.

Why NAT Network is better than NAT for labs: It solves NAT's core limitation by putting multiple VMs on the same network. VMs can discover and reach each other while still having internet access. Suitable for labs that need both VM-to-VM communication and internet (e.g. to pull tools, updates, or run network scans that require name resolution).

Mode 3: Host only

Configuration: Both VMs set to Host-Only Adapter using the VirtualBox Host-Only Ethernet Adapter (192.168.56.0/24), which has its own built-in DHCP server.

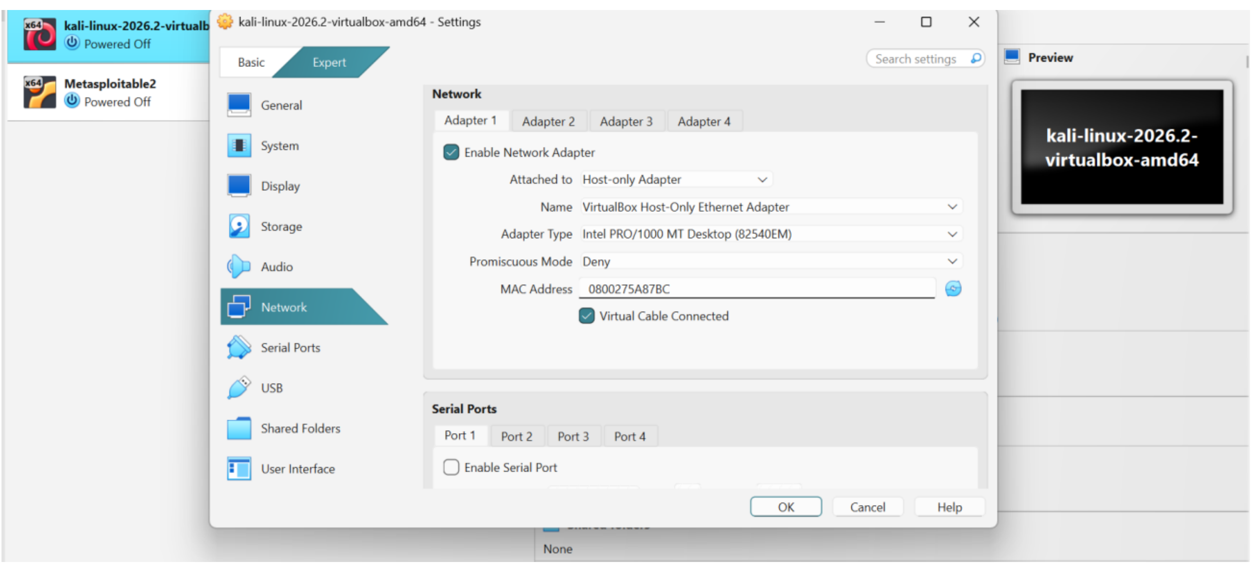
Architecture:

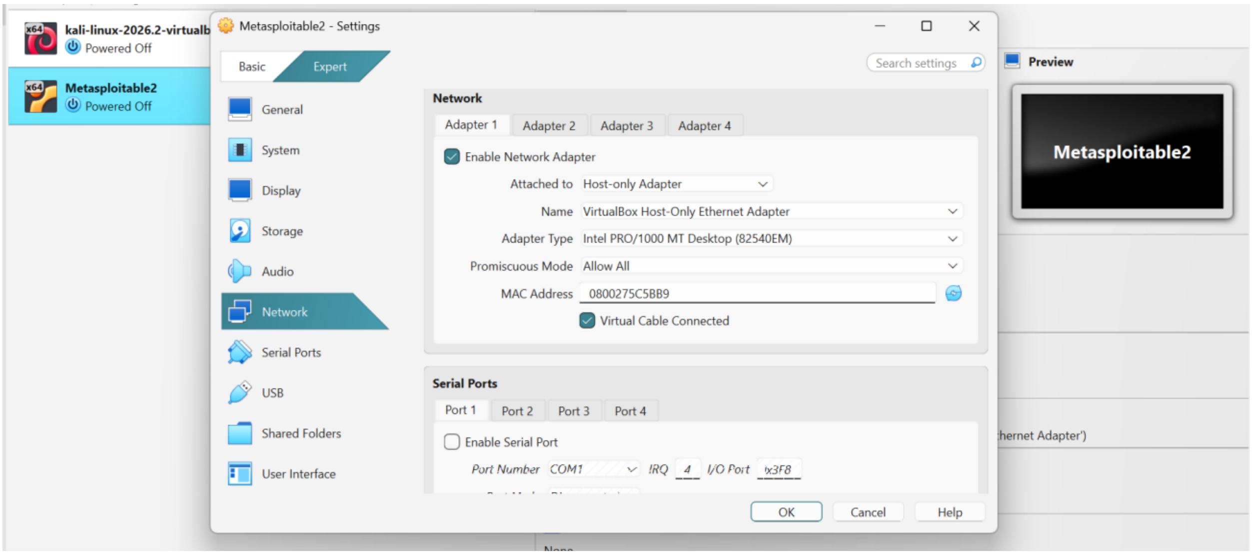
Internet

✗ (no route out)

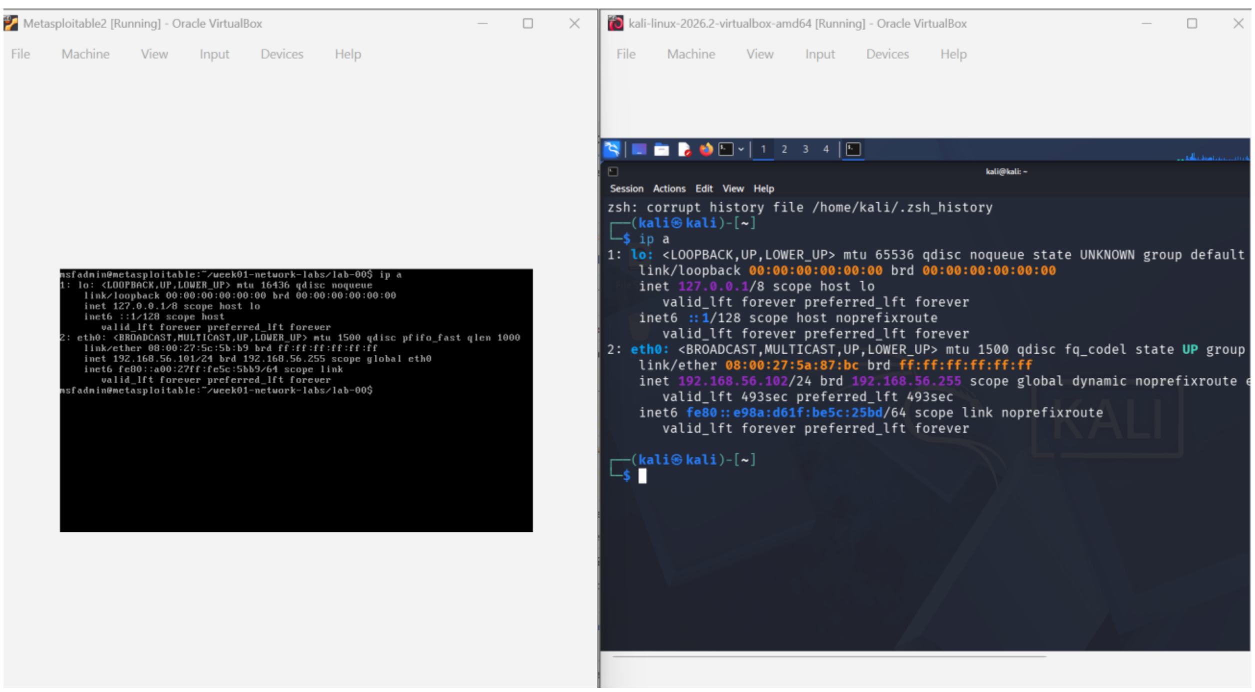
```
Host ——— Host-Only Network (192.168.56.0/24)
        |
        ├── Kali (192.168.56.102)
        |
        └── Metasploitable2 (192.168.56.101)
```

Host can reach both VMs. No internet access.





IP addresses:

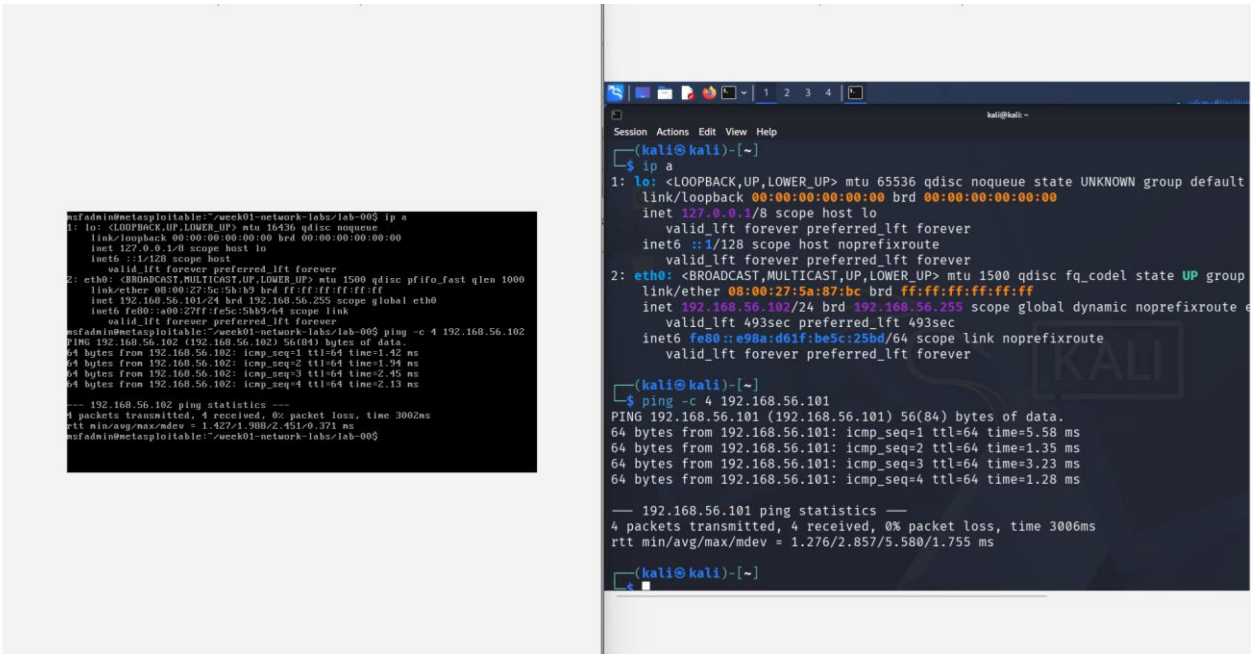


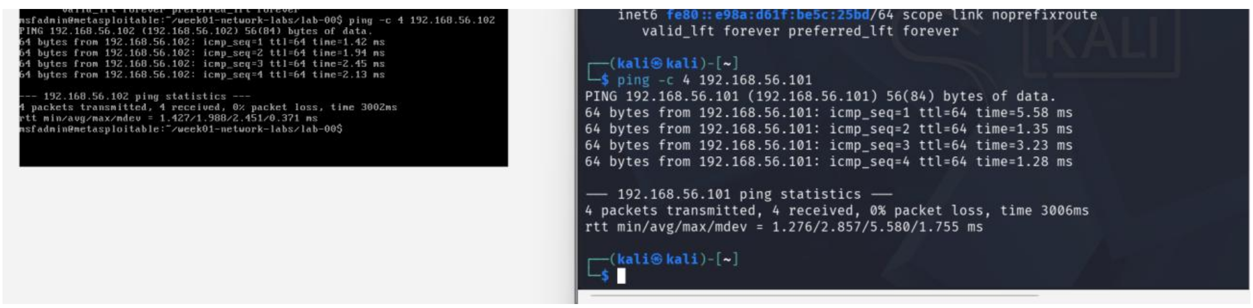
For kali linux: inet :192.168.56.102/24 broadcast address: 192.168.56.255

For metasploitable2: inet: 192.168.56.101/24 broadcast address: 192.168.56.255

Both addresses were assigned automatically by the Host-Only network's built-in DHCP server — no manual configuration needed.

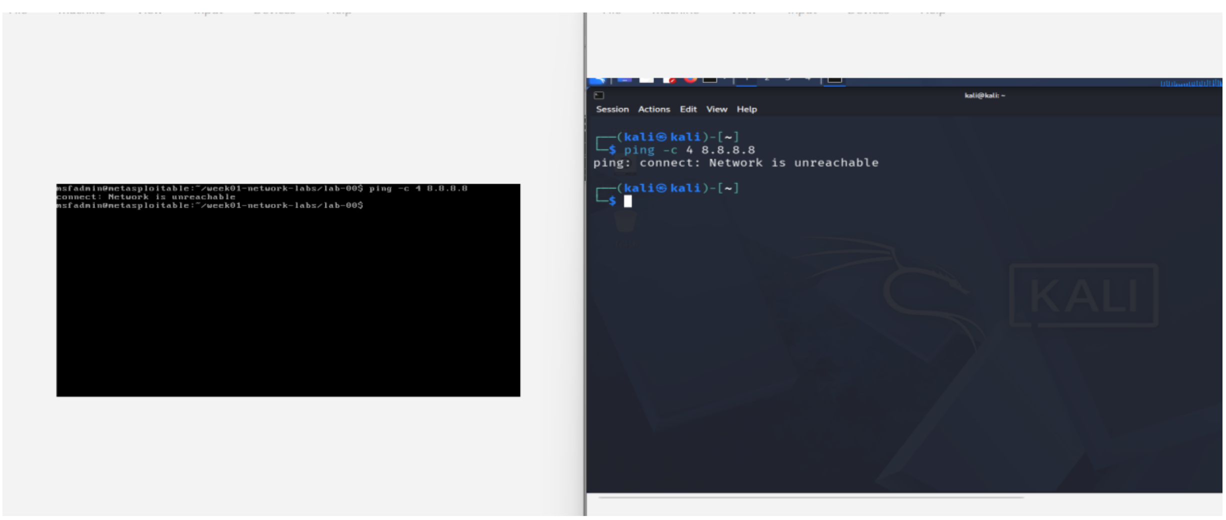
VM-to-VM test:





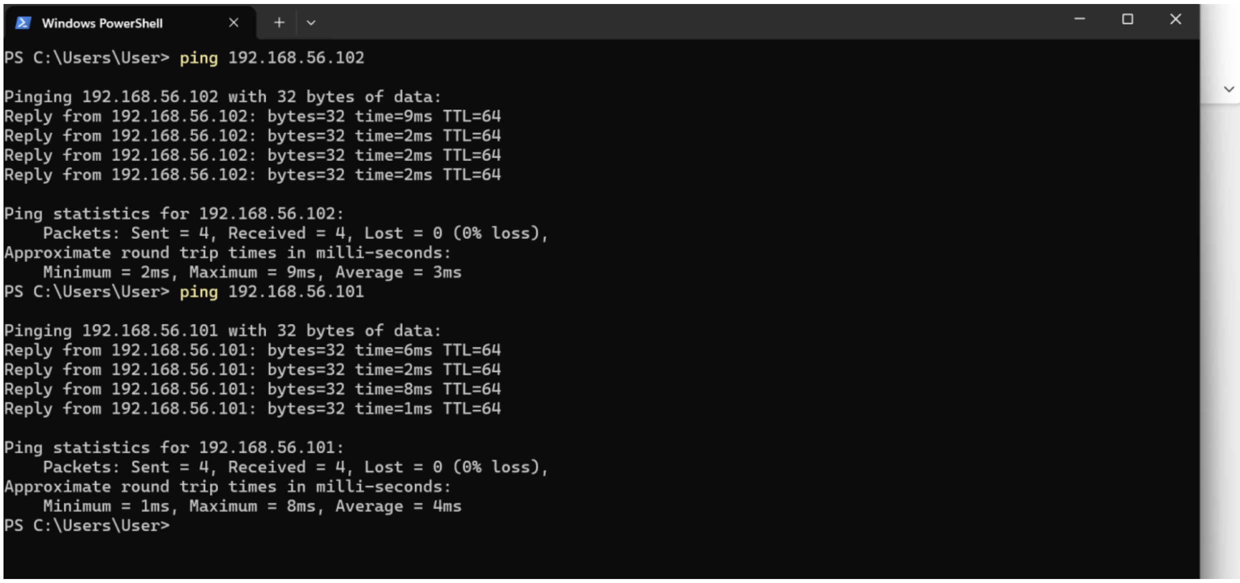
Succeeded in both directions. Both machines are on the same 192.168.56.0/24 subnet, so they can communicate directly.

Internet test:



Failed on both VMs with "Network is unreachable". The Host-Only network is entirely self-contained VirtualBox provides no outbound path to the internet or the physical network. The kernel's routing table has no default gateway to send packets to, so it rejects them before they ever hit the wire.

<u>Host to VM test:</u>



Succeeded from Windows PowerShell to both VMs. This is the defining characteristic of Host-Only mode the Windows host itself has a virtual interface connected to the Host-Only network (you can see it in Windows network adapters as "VirtualBox Host-Only Ethernet Adapter"), and through this interface it can reach all VMs on that network directly.

<u>Why this matters for labs:</u> Host-Only is ideal when you want to run tools from your Windows host machine against the VMs (e.g. connecting to a web service on Metasploitable2 from your browser, or SSH-ing into Kali from your terminal). The host can reach everything, but the vulnerable VM cannot reach the internet and cannot be reached from outside**.**

<u>Mode 4: Internal Network</u>

<u>Configuration:</u> Both VMs set to Internal Network with the same network name (intnet). No DHCP server exists in this mode static IPs were manually assigned using:

sudo ifconfig eth0 192.168.56.10 netmask 255.255.255.0 up (on Kali) sudo ifconfig eth0 192.168.56.20 netmask 255.255.255.0 up (on Metasploitable2)

Architecture:

Internet

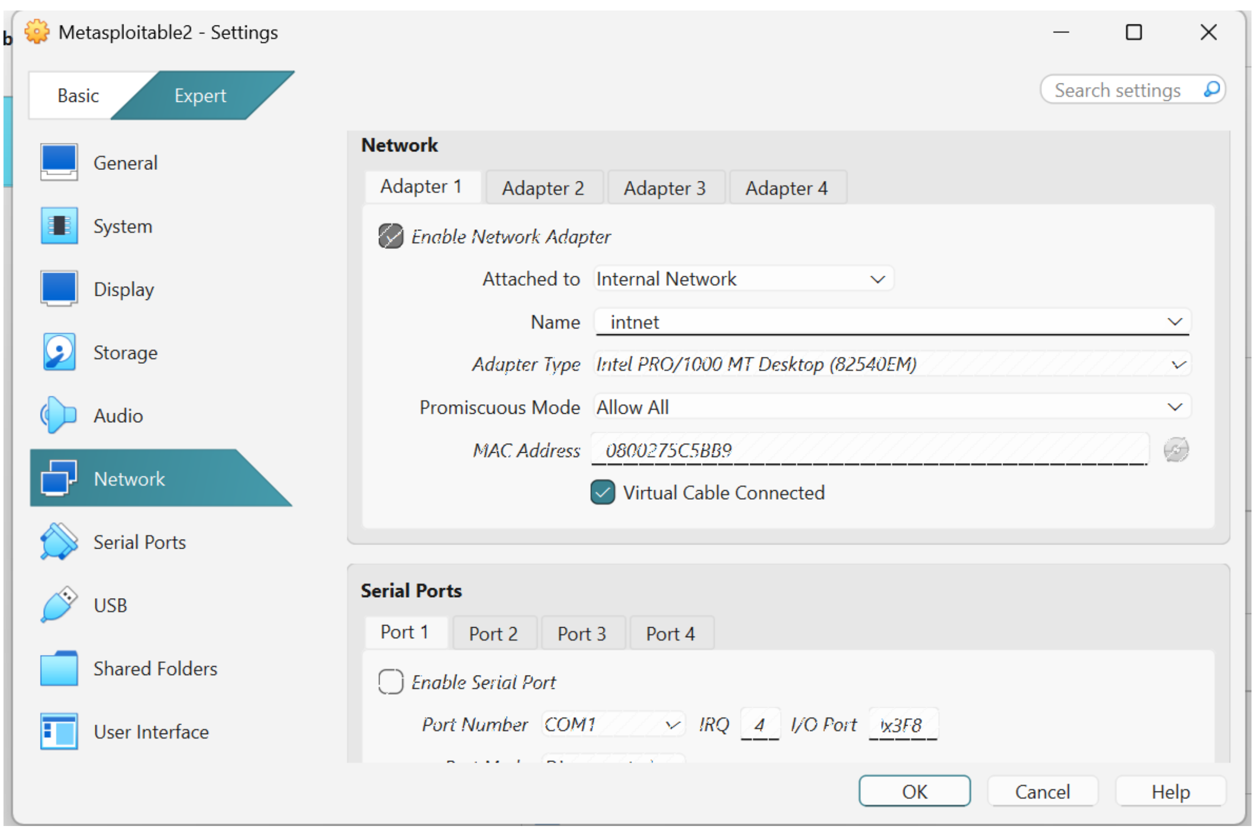
✗ (no route)

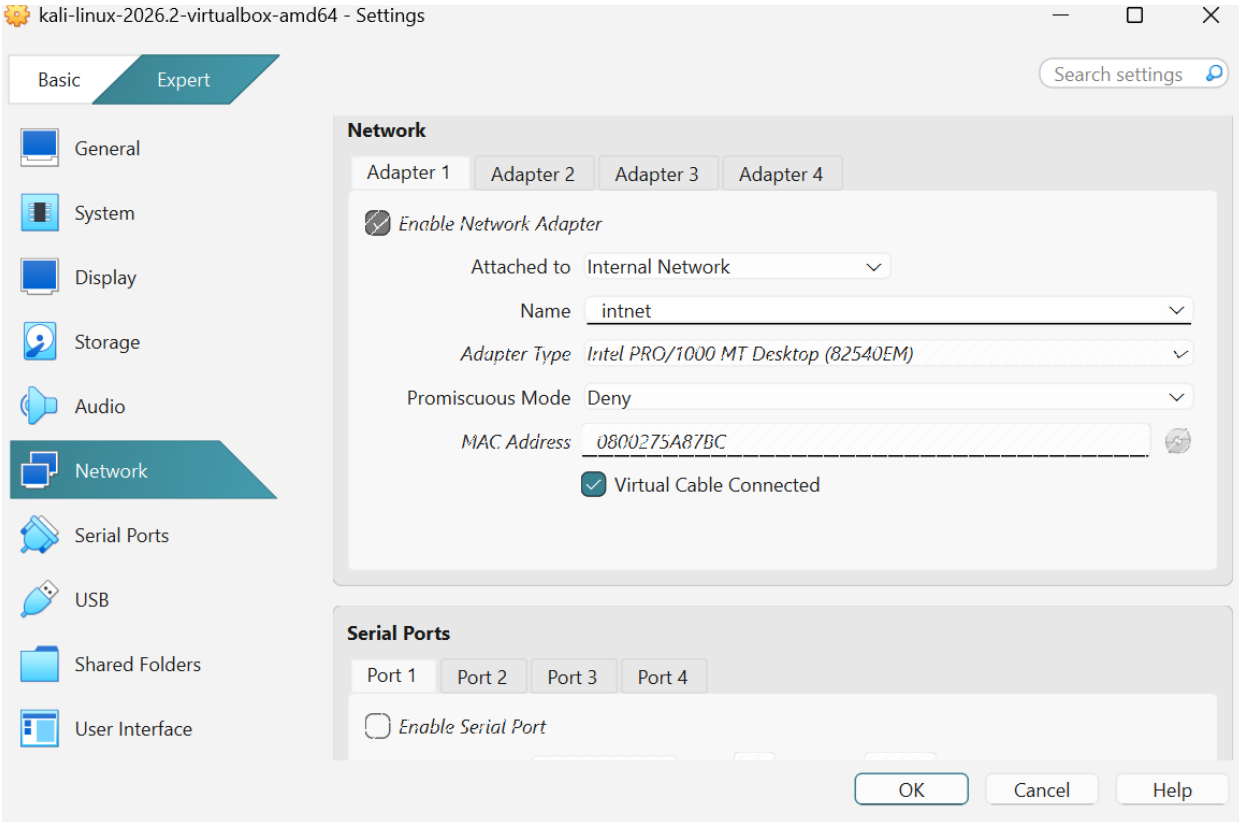
Host

✗ (no route)   Internal Network (intnet)

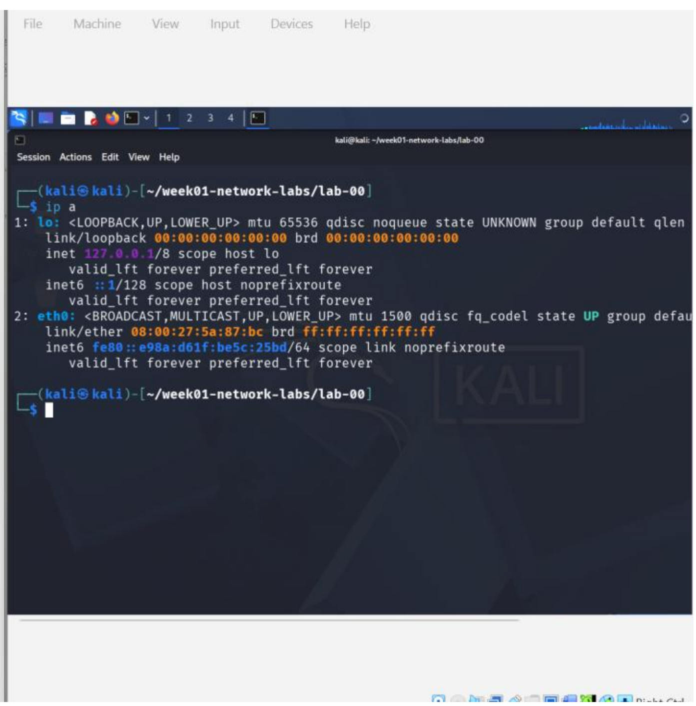
```
                 │
                 ├── Kali (192.168.56.10)
                 │
                 └── Metasploitable2 (192.168.56.20)
```

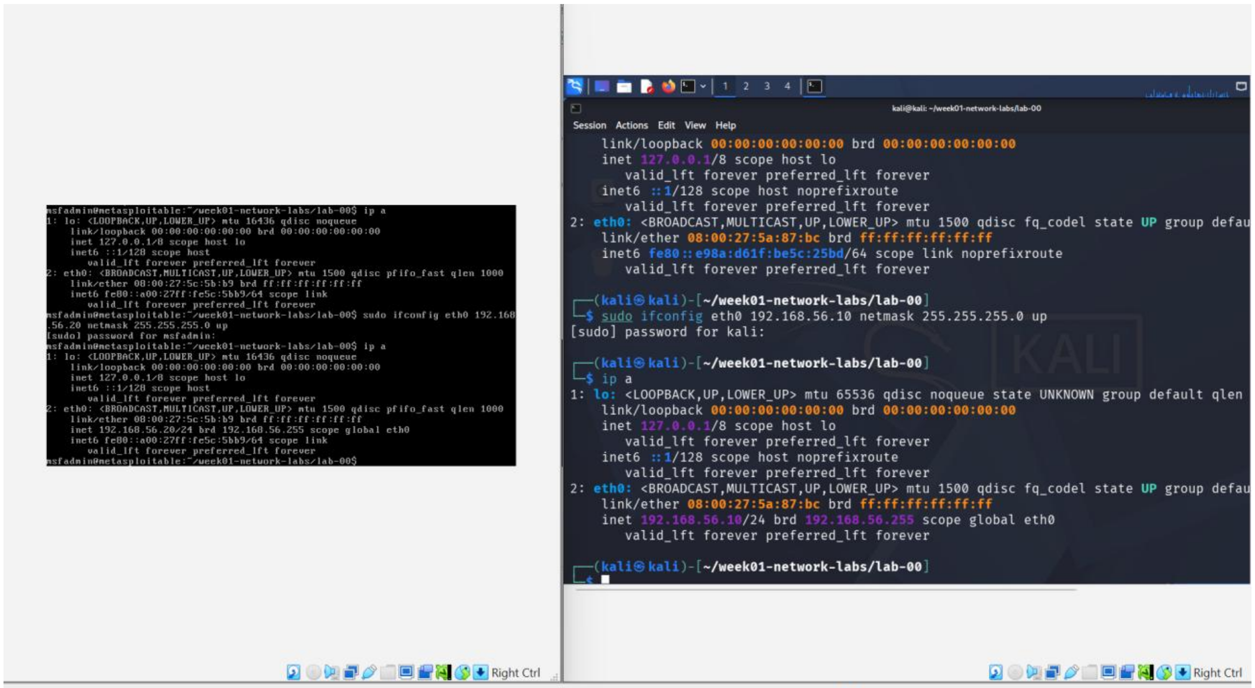
VM-to-VM only. Host and internet completely excluded.





<u>IP addresses:</u>



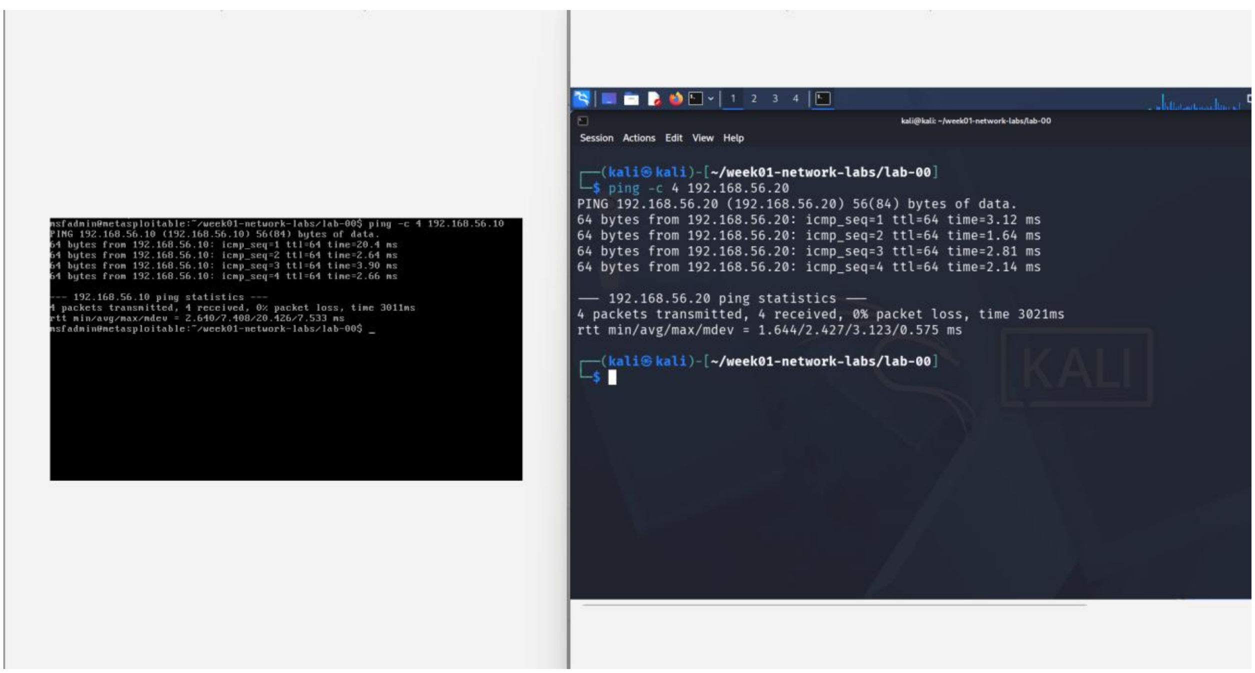


For kali linux: inet:192.168.56.10 broadcast address:192.168.56.255

For metasploitable2: inet:192.168.56.20 broadcast address:192.168.56.255

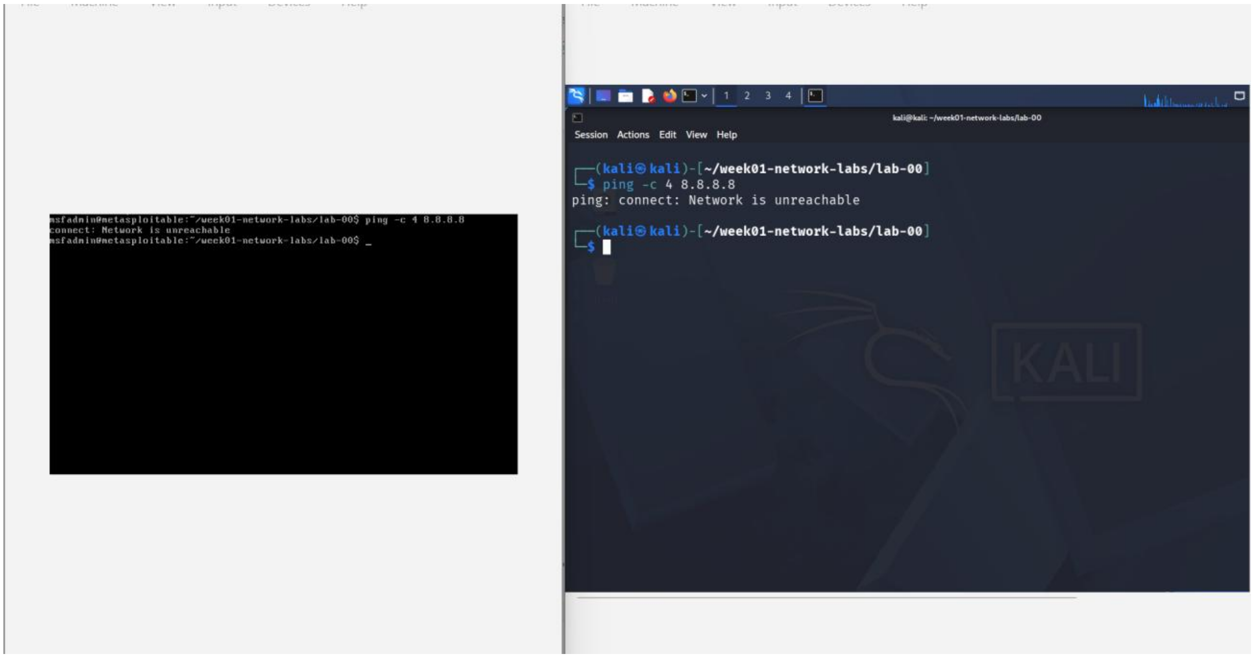
When both VMs first booted with no IP configured, Metasploitable2 showed no IPv4 address at all on eth0 — only the link-local address. This is expected: Internal Network has no DHCP server by design. Static assignment is mandatory.

VM-to-VM test:



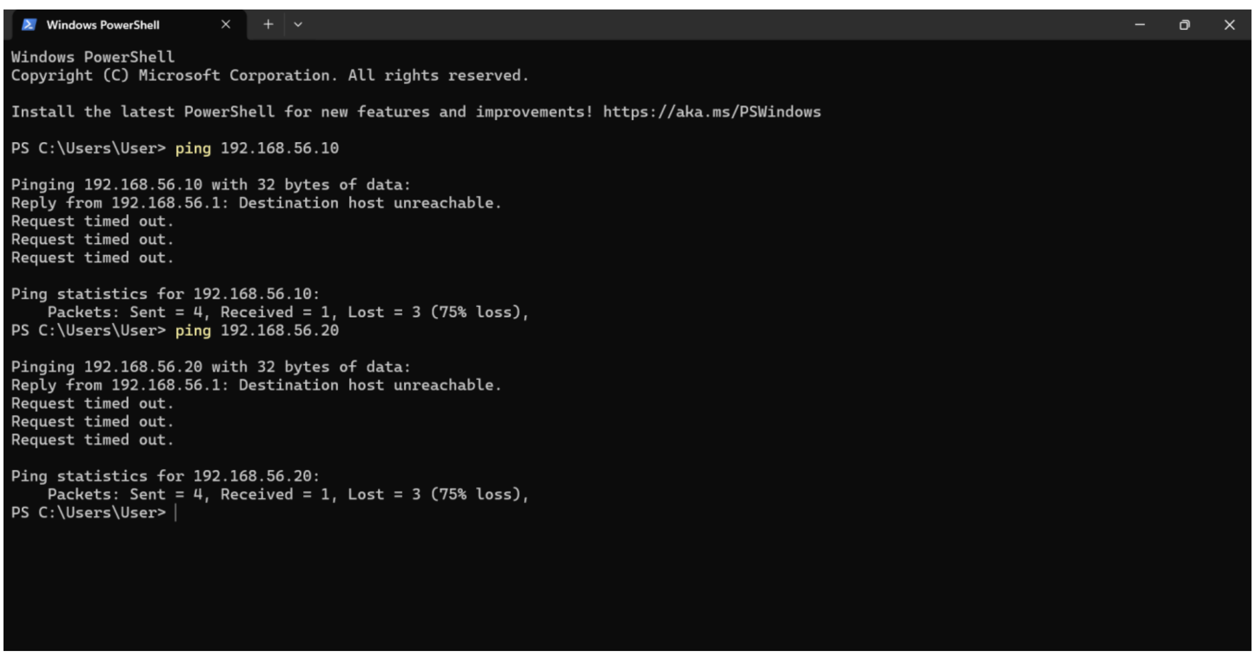
Succeeded after manual IP assignment. Both VMs are on the same subnet and can communicate directly.

Internet test:



Failed with "Network is unreachable" same reason as Host-Only, no gateway exists.

Host to VM test:



Failed with "Destination host unreachable" from PowerShell. The error text is slightly different from the VM-side error and the distinction is important: when Windows tried to ping

192.168.56.10, its own routing table had no idea how to reach that subnet. The ping request was forwarded to Windows' default gateway (your real router), which responded with "Destination host unreachable" because no such device exists on the real network. The VM never received the ping at all. This is completely different from Host-Only, where the host has a dedicated virtual interface into the lab network.

Why Internal Network is the strictest mode: Nothing can enter or exit except between VMs on the same named internal network. This is the most appropriate mode for controlled security experiments where you want zero risk of traffic leaking to or from the real network. The trade-off is manual IP management and no host access.

Why we use this mode for the packet capture, ARP, and scanning labs: It gives completely clean traffic every packet you see in Wireshark is yours, generated by your lab. No background noise from the host's real network.

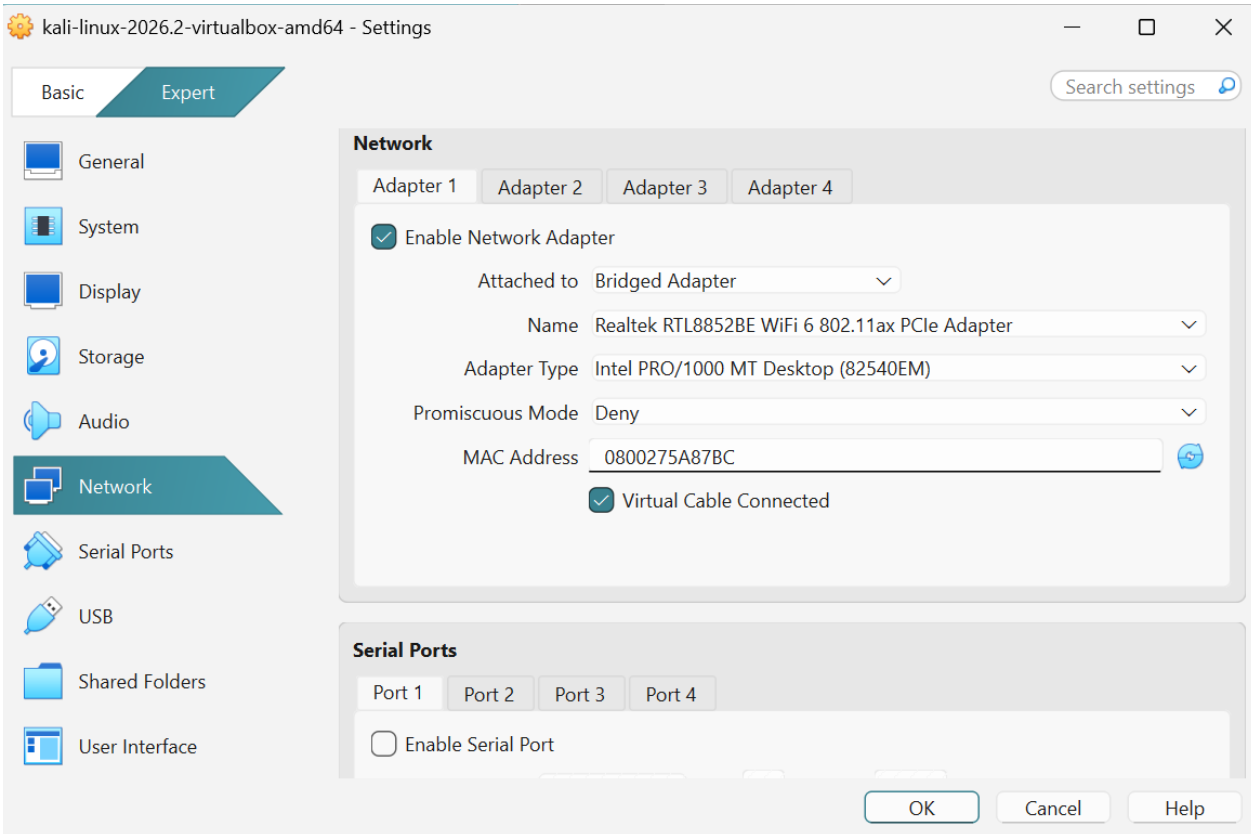
Mode 5:Bridged

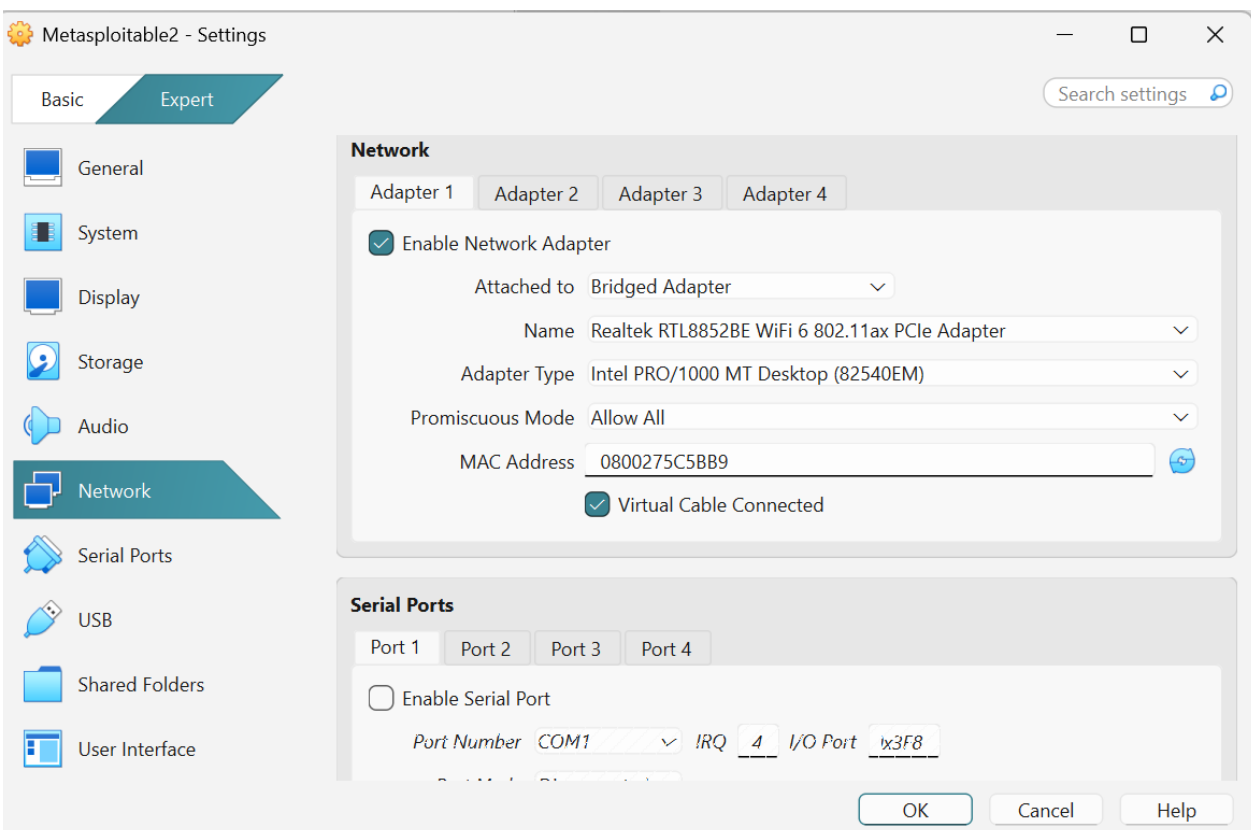
Configuration: Both VMs set to Bridged Adapter, connected to the host's physical WiFi adapter (Realtek RTL8852BE).

Architecture:

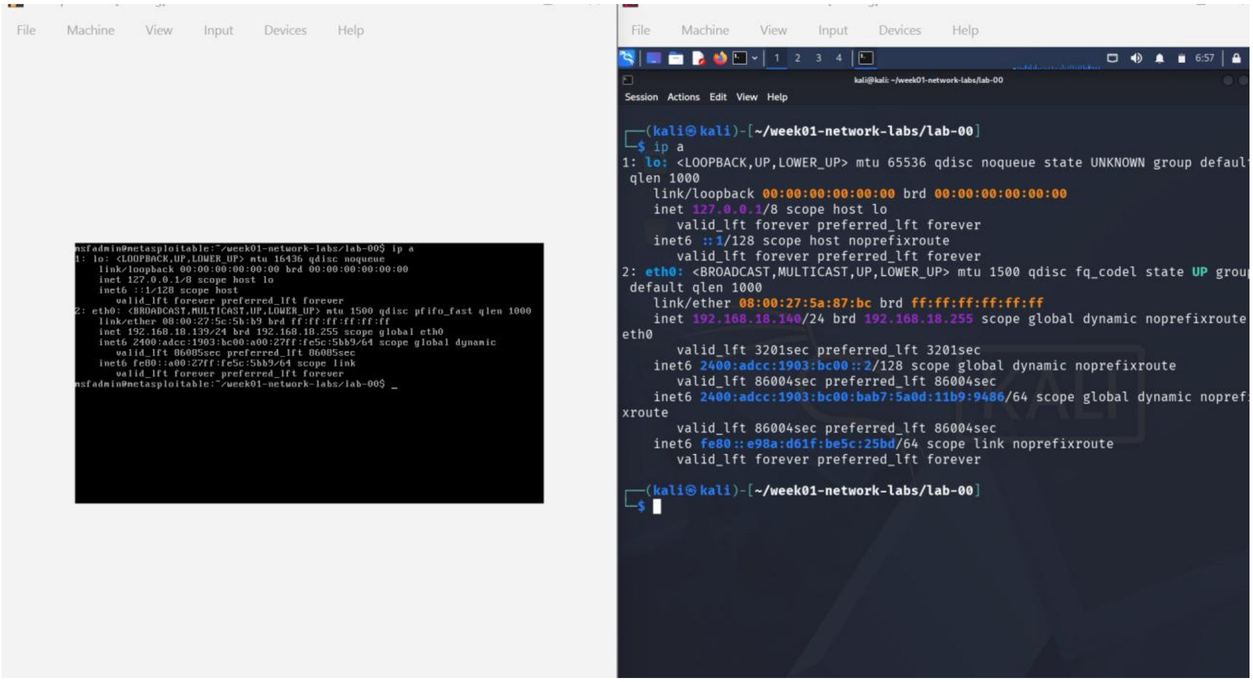
```
Physical Router / Internet
        |
Physical LAN (192.168.18.0/24)
        |
        ├── Host (192.168.18.x)
        |
        ├── Kali (192.168.18.140)
        |
        └── Metasploitable2 (192.168.18.139)
```

All devices are full peers on the real network


kali-linux-2026.2-virtualbox-amd64 - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to Bridged Adapter
Name Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
Adapter Type Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode Deny
MAC Address 0800275A87BC
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
OK
Cancel
Help


Metasploitable2 - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to Bridged Adapter
Name Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
Adapter Type Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode Allow All
MAC Address 0800275C5BB9
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
Port Number COM1
IRQ 4
I/O Port 0x3F8
OK
Cancel
Help

IP addresses:

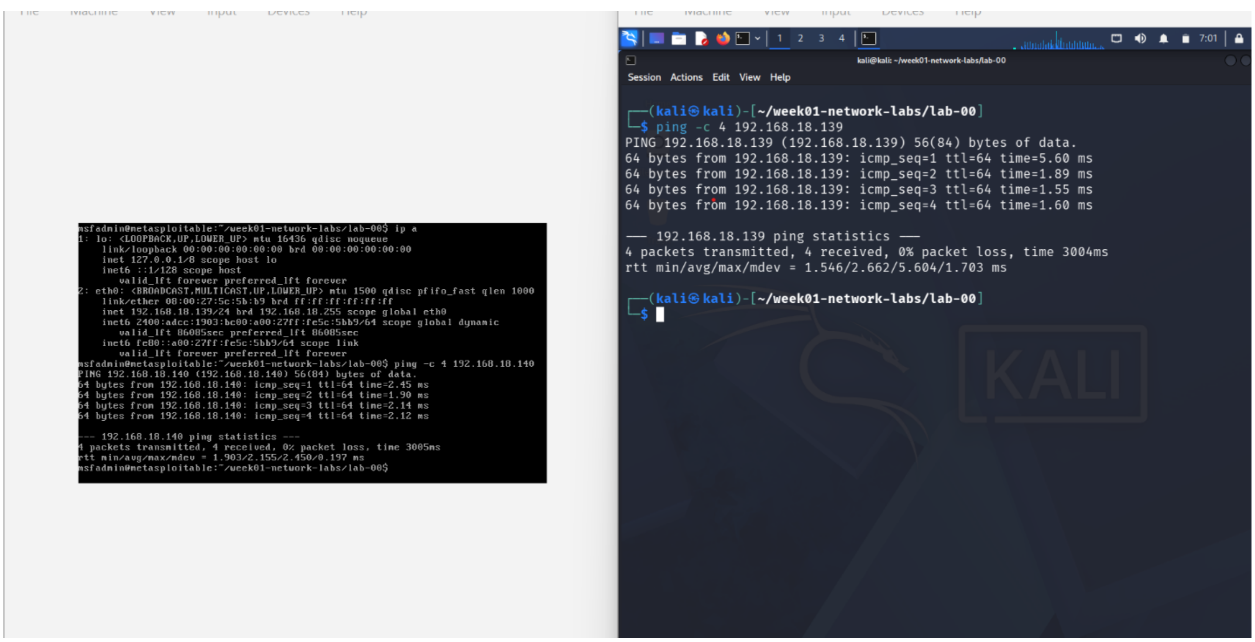


For kali linux: inet: 192.168.18.140 broadcast address: 192.168.18.255

For metasploitable2: inet: 192.168.18.139 broadcast address : 192.168.18.255

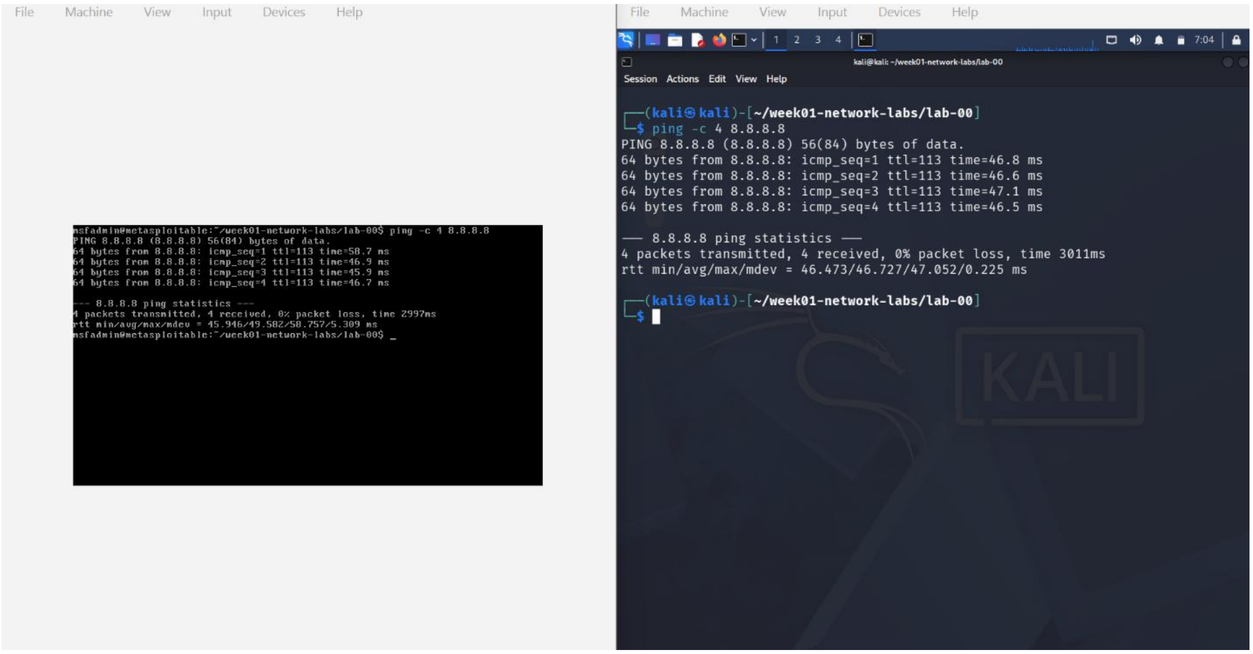
Both addresses were assigned by the physical router's DHCP server — the same DHCP server that assigns addresses to your phone, laptop, and every other real device on your WiFi. The VMs appear to the router as ordinary devices with their own MAC addresses.

VM-to-VM test:



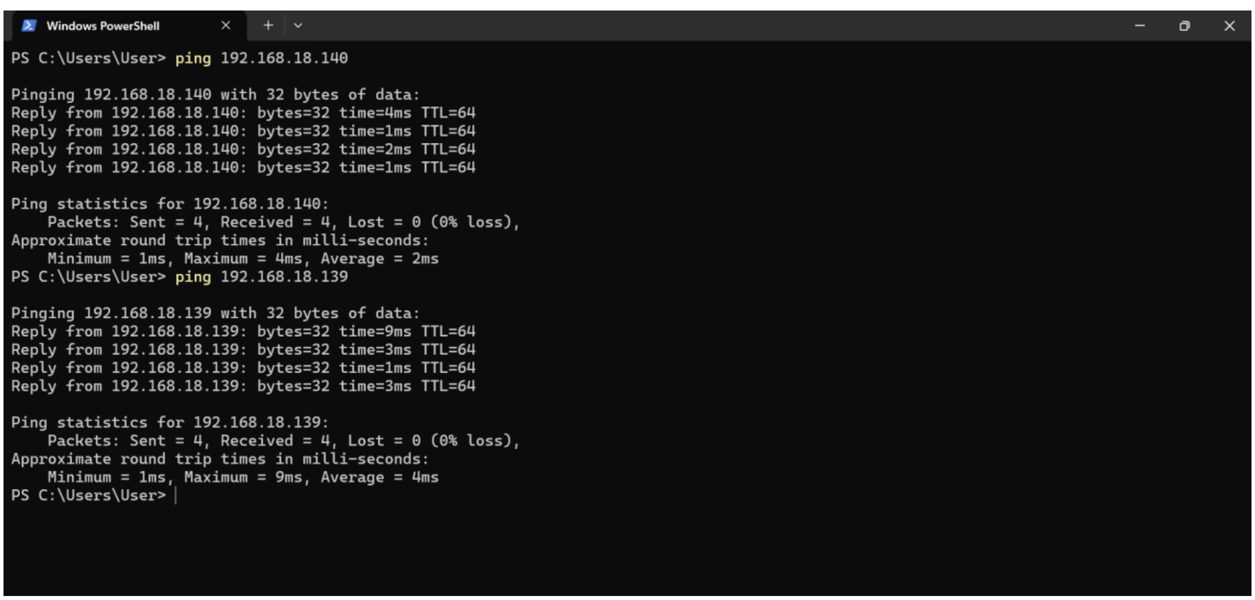
Succeeded both VMs are on the same physical LAN.

Internet test:



Succeeded with noticeably higher latency (40–100ms) compared to previous modes. This is because packets now travel through the real physical router to reach the internet, rather than through VirtualBox's internal NAT engine. The higher latency is real-world network latency.

Host to VM test:



Tested, and inherently succeeds all three devices (host, Kali, Metasploitable2) are peers on the same physical LAN and can reach each other directly.

Critical security warning: Metasploitable2 is intentionally vulnerable. In Bridged mode it is directly visible to every device on your real network your phone, other computers, your router. Anyone or anything on that network could potentially interact with it. Never leave Metasploitable2 in Bridged mode unattended or on a shared/public network.

6. Overall Comparison Table

Mode	VM-to-VM	Internet	Host → VM	IP Assignment
NAT	✗ (self-loop)	✓	✗	Automatic, identical per VM
NAT Network	✓	✓	✗	Automatic, shared pool
Host-Only	✓	✗	✓	Automatic (DHCP built-in)
Internal Network	✓ (after manual config)	✗	✗	Manual (no DHCP)
Bridged	✓	✓	✓ (real LAN peer)	Automatic, real router DHCP

7. Conclusion and Decision Making Based on Labs

This investigation showed that network mode selection is a foundational design decision, not a cosmetic setting. The wrong choice can make an entire lab impossible (NAT for a multi-machine attack) or introduce real risk (Bridged with Metasploitable2 on shared WiFi).

For future labs, the selection guide is:

- **Internal Network**: all packet capture, ARP, ICMP, TCP, DNS, and scanning labs. Clean isolated traffic, no noise, no risk of exposure. This is the default for all offensive labs in this curriculum.
- **NAT Network**: any lab that needs VM-to-VM communication AND internet access simultaneously (e.g. pulling a tool on Kali while also running scans against Metasploitable2).
- **Host-Only**: any lab where you want to connect to a VM service from your Windows host (e.g. opening Metasploitable2's web interface in your Windows browser, or SSH-ing into Kali from Windows Terminal).

- **NAT**: single-VM labs only, or when a VM needs internet access with no interaction with other VMs.
- **Bridged**: use only when explicitly required and only briefly. Always shut down Metasploitable2 before switching it to Bridged.